

Guitar Tutor

EECS 373 · INTRODUCTION TO EMBEDDED SYSTEMS

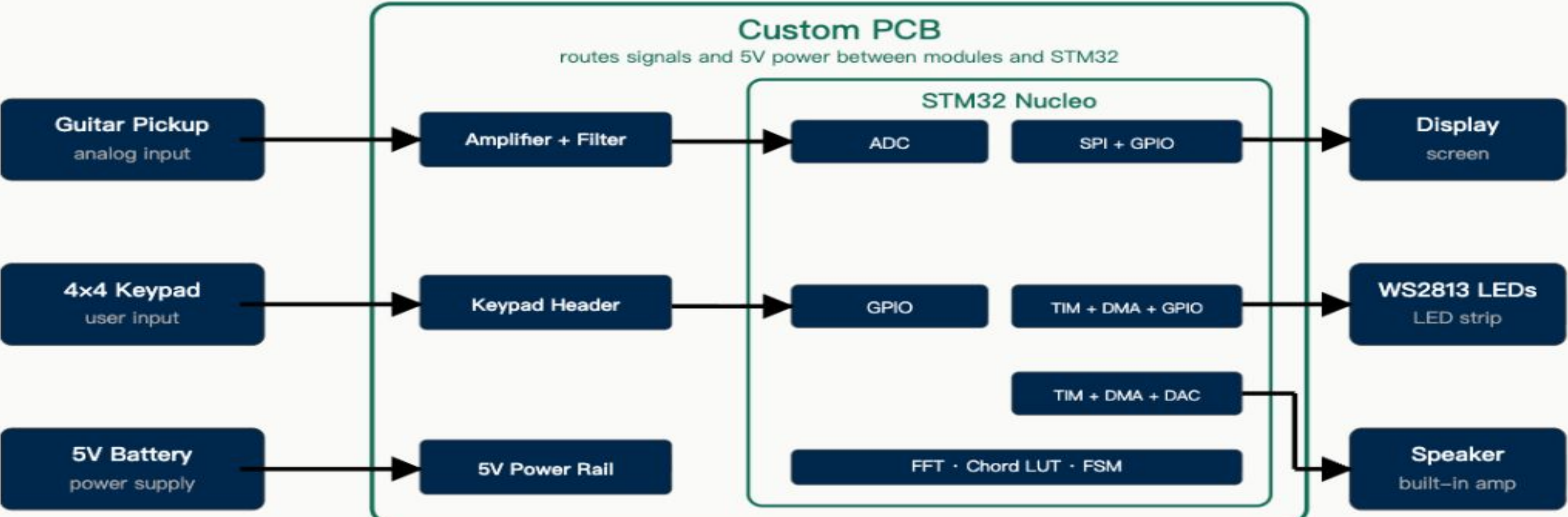
XIAOCAN SHI · XINYU HU · LINGXIAO YANG
WENLAN WU · RUI JIANG



MOTIVATION

- No visual guidance in traditional guitar learning
- Chord shapes are hard to memorize without instant feedback
- Existing tools are fixed to one instrument — not portable

SYSTEM ARCHITECTURE



External modules connect through the custom PCB; STM32 peripherals drive each signal path.

The guitar pickup signal is conditioned by the amplifier/filter AFE before entering the STM32 ADC. The keypad uses row/column GPIO scan lines. The display uses SPI, the WS2813 LEDs use a timed LED data signal, and the speaker receives DAC audio output through its built-in amplifier.

KEY FEATURES

- LED chord projection**
Addressable RGB per fret + string
- 512-pt FFT verification**
Detects chord from mic, confirms accuracy
- 320x480 TFT display**
Chord name, dot positions, progression
- Training mode FSM**
Auto-cycles via timed interrupt
- Portable demo board**
Standalone — no guitar needed

DISPLAY SCREENS

The display screens show the following content:

- Chord view:** Fret dot positions on a guitar fretboard.
- Mode select:** Manual menu with options: MENU, Chord, Exit. Instructions: Press 'B/C' to go next/prev, Press 'D' to enter.
- Audio Process:** FFT Spectrum showing frequency response.
- Test Feedback:** Pass / fail result. Example: Detected Chord: C, Score: 96.